

Task- clause

1. Fetch all employees who earn more than 40,000.
2. Get all employees whose names start with the letter 'A'.
3. Retrieve all employees where the department is either 'HR' or 'Finance'.
4. Find all employees where the manager field has no value (is NULL).
5. Count the number of employees in each department, but only include employees earning more than 30,000.
6. Find the average salary for each job_role in the 'IT' department.
7. Show the total salary paid per department and per job_role for employees aged under 35.
8. Find the total salary paid to employees in each department.
9. Count the total number of employees in each department.
10. Find the average salary of employees in each department.
11. Show the count of employees for each department and job_role combination.

```
SELECT * FROM employees;
```

Empty set (0.0111 sec)

```
MySQL localhost:33060+ ssl mydata SQL > INSERT INTO employees (name, age, department, job_role, salary, manager)
```

```
                                -> VALUES
                                -> ('Anil', 28, 'IT', 'Developer', 45000,
'Ravi'),
                                -> ('Arjun', 32, 'IT', 'Tester', 38000,
'Ravi'),
                                -> ('Rahul', 29, 'HR', 'HR Executive',
42000, 'Priya'),
                                -> ('Sneha', 35, 'Finance', 'Accountant',
50000, 'Kiran'),
                                -> ('Asha', 26, 'Finance', 'Analyst',
36000, 'Kiran'),
                                -> ('Priya', 31, 'IT', 'Developer', 55000,
'Ravi'),
                                -> ('Kiran', 40, 'HR', 'Manager', 65000,
NULL),
                                -> ('Meena', 30, 'Sales', 'Sales
Executive', 32000, 'Kiran'),
                                -> ('Ajay', 27, 'IT', 'Developer', 48000,
```

```
'Ravi'),
                                -> ('Divya', 33, 'Finance', 'Analyst',
41000, 'Kiran');
```

ERROR: 1054: Unknown column 'name' in 'field list'

```
MySQL localhost:33060+ ssl mydata SQL > DESCRIBE employees;
```

Field	Type	Null	Key	Default	Extra
emp_id	varchar(50)	YES		NULL	
emp_name	varchar(30)	YES		NULL	
salary	decimal(10,2)	YES		NULL	

3 rows in set (0.0040 sec)

```
MySQL localhost:33060+ ssl mydata SQL > INSERT INTO employees (emp_id,
emp_name, salary)
```

```
-> VALUES
-> ('E101', 'Anil', 45000),
-> ('E102', 'Arjun', 38000),
-> ('E103', 'Rahul', 42000),
-> ('E104', 'Sneha', 50000),
-> ('E105', 'Asha', 36000),
-> ('E106', 'Priya', 55000),
-> ('E107', 'Kiran', 65000),
-> ('E108', 'Meena', 32000),
-> ('E109', 'Ajay', 48000),
-> ('E110', 'Divya', 41000);
```

Query OK, 10 rows affected (0.0157 sec)

Records: 10 Duplicates: 0 Warnings: 0

```
MySQL localhost:33060+ ssl mydata SQL > SELECT * FROM employees;
```

emp_id	emp_name	salary
E101	Anil	45000.00
E102	Arjun	38000.00
E103	Rahul	42000.00
E104	Sneha	50000.00
E105	Asha	36000.00
E106	Priya	55000.00
E107	Kiran	65000.00
E108	Meena	32000.00
E109	Ajay	48000.00
E110	Divya	41000.00

10 rows in set (0.0018 sec)

```
MySQL localhost:33060+ ssl mydata SQL > SELECT *
```

```
-> FROM employees
-> WHERE salary > 40000;
```

emp_id	emp_name	salary
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E101	Anil	45000.00
E103	Rahul	42000.00
E104	Sneha	50000.00
E106	Priya	55000.00
E107	Kiran	65000.00
E109	Ajay	48000.00
E110	Divya	41000.00

7 rows in set (0.0037 sec)

```
MySQL localhost:33060+ ssl mydata SQL > SELECT * FROM employees
-> WHERE name LIKE '%A';
```

ERROR: 1054: Unknown column 'name' in 'where clause'

```
MySQL localhost:33060+ ssl mydata SQL > SELECT *
-> FROM employees
-> WHERE name LIKE 'A%';
```

ERROR: 1054: Unknown column 'name' in 'where clause'

```
MySQL localhost:33060+ ssl mydata SQL > SELECT *
-> FROM employees
-> WHERE emp_name LIKE 'A%';
```

emp_id	emp_name	salary
E101	Anil	45000.00
E102	Arjun	38000.00
E105	Asha	36000.00
E109	Ajay	48000.00

4 rows in set (0.0012 sec)

```
MySQL localhost:33060+ ssl mydata SQL > SELECT *
-> FROM employees
-> WHERE department IN ('HR', 'Finance');
```

ERROR: 1054: Unknown column 'department' in 'where clause'

```
MySQL localhost:33060+ ssl mydata SQL > ALTER TABLE employees
-> ADD COLUMN age INT,
-> ADD COLUMN department VARCHAR(50),
-> ADD COLUMN job_role VARCHAR(50),
-> ADD COLUMN manager VARCHAR(50);
```

Query OK, 0 rows affected (0.0334 sec)

Records: 0 Duplicates: 0 Warnings: 0

```
MySQL localhost:33060+ ssl mydata SQL > DESCRIBE employees;
```

Field	Type	Null	Key	Default	Extra
emp_id	varchar(50)	YES		NULL	
emp_name	varchar(30)	YES		NULL	
salary	decimal(10,2)	YES		NULL	
age	int	YES		NULL	
department	varchar(50)	YES		NULL	

job_role	varchar(50)	YES		NULL	
manager	varchar(50)	YES		NULL	

7 rows in set (0.0036 sec)

```
MySQL localhost:33060+ ssl mydata SQL > UPDATE employees
-> SET age = 28, department = 'IT',
job_role = 'Developer', manager = 'Ravi'
-> WHERE emp_id = 'E101';
```

Query OK, 1 row affected (0.0059 sec)

Rows matched: 1 Changed: 1 Warnings: 0

```
MySQL localhost:33060+ ssl mydata SQL >
MySQL localhost:33060+ ssl mydata SQL > UPDATE employees
-> SET age = 32, department = 'IT',
job_role = 'Tester', manager = 'Ravi'
-> WHERE emp_id = 'E102';
```

Query OK, 1 row affected (0.0044 sec)

Rows matched: 1 Changed: 1 Warnings: 0

```
MySQL localhost:33060+ ssl mydata SQL >
MySQL localhost:33060+ ssl mydata SQL > UPDATE employees
-> SET age = 29, department = 'HR',
job_role = 'HR Executive', manager = 'Priya'
-> WHERE emp_id = 'E103';
```

Query OK, 1 row affected (0.0040 sec)

Rows matched: 1 Changed: 1 Warnings: 0

```
MySQL localhost:33060+ ssl mydata SQL >
MySQL localhost:33060+ ssl mydata SQL > UPDATE employees
-> SET age = 35, department = 'Finance',
job_role = 'Accountant', manager = 'Kiran'
-> WHERE emp_id = 'E104';
```

Query OK, 1 row affected (0.0048 sec)

Rows matched: 1 Changed: 1 Warnings: 0

```
MySQL localhost:33060+ ssl mydata SQL >
MySQL localhost:33060+ ssl mydata SQL > UPDATE employees
-> SET age = 26, department = 'Finance',
job_role = 'Analyst', manager = 'Kiran'
-> WHERE emp_id = 'E105';
```

Query OK, 1 row affected (0.0041 sec)

Rows matched: 1 Changed: 1 Warnings: 0

```
MySQL localhost:33060+ ssl mydata SQL >
MySQL localhost:33060+ ssl mydata SQL > UPDATE employees
-> SET age = 31, department = 'IT',
job_role = 'Developer', manager = 'Ravi'
-> WHERE emp_id = 'E106';
```

Query OK, 1 row affected (0.0043 sec)

```

Rows matched: 1  Changed: 1  Warnings: 0
MySQL localhost:33060+ ssl mydata SQL >
MySQL localhost:33060+ ssl mydata SQL > UPDATE employees
-> SET age = 40, department = 'HR',
job_role = 'Manager', manager = NULL
-> WHERE emp_id = 'E107';
Query OK, 1 row affected (0.0052 sec)

```

```

Rows matched: 1  Changed: 1  Warnings: 0
MySQL localhost:33060+ ssl mydata SQL >
MySQL localhost:33060+ ssl mydata SQL > UPDATE employees
-> SET age = 30, department = 'Sales',
job_role = 'Sales Executive', manager = 'Kiran'
-> WHERE emp_id = 'E108';
Query OK, 1 row affected (0.0037 sec)

```

```

Rows matched: 1  Changed: 1  Warnings: 0
MySQL localhost:33060+ ssl mydata SQL >
MySQL localhost:33060+ ssl mydata SQL > UPDATE employees
-> SET age = 27, department = 'IT',
job_role = 'Developer', manager = 'Ravi'
-> WHERE emp_id = 'E109';
Query OK, 1 row affected (0.0041 sec)

```

```

Rows matched: 1  Changed: 1  Warnings: 0
MySQL localhost:33060+ ssl mydata SQL >
MySQL localhost:33060+ ssl mydata SQL > UPDATE employees
-> SET age = 33, department = 'Finance',
job_role = 'Analyst', manager = 'Kiran'
-> WHERE emp_id = 'E110';
Query OK, 1 row affected (0.0048 sec)

```

```

Rows matched: 1  Changed: 1  Warnings: 0
MySQL localhost:33060+ ssl mydata SQL > SELECT *
-> FROM employees
-> WHERE department IN ('HR', 'Finance');

```

emp_id	emp_name	salary	age	department	job_role	manager
E103	Rahul	42000.00	29	HR	HR Executive	Priya
E104	Sneha	50000.00	35	Finance	Accountant	Kiran
E105	Asha	36000.00	26	Finance	Analyst	Kiran
E107	Kiran	65000.00	40	HR	Manager	NULL
E110	Divya	41000.00	33	Finance	Analyst	Kiran

5 rows in set (0.0014 sec)

```

MySQL localhost:33060+ ssl mydata SQL > SELECT *
-> FROM employees
-> WHERE manager IS NULL;

```

```

+-----+-----+-----+-----+-----+-----+-----+

```

emp_id	emp_name	salary	age	department	job_role	manager
E107	Kiran	65000.00	40	HR	Manager	NULL

1 row in set (0.0012 sec)

MySQL localhost:33060+ ssl mydata SQL > SELECT department, COUNT(*) AS employee_count

-> FROM employees
-> WHERE salary > 30000
-> GROUP BY department;

department	employee_count
IT	4
HR	2
Finance	3
Sales	1

4 rows in set (0.0015 sec)

MySQL localhost:33060+ ssl mydata SQL > SELECT job_role, AVG(salary) AS average_salary

-> FROM employees
-> WHERE department = 'IT'
-> GROUP BY job_role;

job_role	average_salary
Developer	49333.333333
Tester	38000.000000

2 rows in set (0.0024 sec)

MySQL localhost:33060+ ssl mydata SQL > SELECT department, job_role, SUM(salary) AS total_salary

-> FROM employees
-> WHERE age < 35
-> GROUP BY department, job_role;

department	job_role	total_salary
IT	Developer	148000.00
IT	Tester	38000.00
HR	HR Executive	42000.00
Finance	Analyst	77000.00
Sales	Sales Executive	32000.00

5 rows in set (0.0017 sec)

MySQL localhost:33060+ ssl mydata SQL > SELECT department, SUM(salary) AS total_salary

-> FROM employees
-> GROUP BY department;

department	total_salary
IT	186000.00
HR	107000.00
Finance	127000.00
Sales	32000.00

4 rows in set (0.0013 sec)

MySQL localhost:33060+ ssl mydata SQL > SELECT department, COUNT(*) AS employee_count

-> FROM employees

-> GROUP BY department;

department	employee_count
IT	4
HR	2
Finance	3
Sales	1

4 rows in set (0.0015 sec)

MySQL localhost:33060+ ssl mydata SQL > SELECT department, AVG(salary) AS average_salary

-> FROM employees

-> GROUP BY department;

department	average_salary
IT	46500.000000
HR	53500.000000
Finance	42333.333333
Sales	32000.000000

4 rows in set (0.0016 sec)

MySQL localhost:33060+ ssl mydata SQL > SELECT department, job_role, COUNT(*) AS employee_count

-> FROM employees

-> GROUP BY department, job_role;

department	job_role	employee_count
IT	Developer	3
IT	Tester	1
HR	HR Executive	1
Finance	Accountant	1
Finance	Analyst	2
HR	Manager	1
Sales	Sales Executive	1

7 rows in set (0.0014 sec)

MySQL localhost:33060+ ssl mydata SQL >